

Pharmacy E-Commerce with Prescription Validation

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Abstract



Pharmacy E-Commerce with Prescription Validation is a full-stack web application enabling secure online medicine purchases. Customers can browse, search, and order medicines — with OTC medicines available directly and prescription-only medicines requiring pharmacist verification.

A centralised **MySQL database** manages medicines, inventory, prescriptions, orders, and payments. Secure authentication and prescription validation prevent unauthorised purchases, demonstrating practical implementation of **database design, SQL, APIs, and full-stack development**.



Problem Statement

Unverified Prescription Sales

Many existing platforms lack secure mechanisms to verify prescriptions before dispensing prescription-only medicines, risking misuse and legal non-compliance.

Slow Manual Processes

Manual prescription verification is time-consuming and inconvenient for both customers and pharmacists, creating bottlenecks in order fulfilment.

Fragmented Systems

Disconnected inventory, customer, and prescription records reduce operational efficiency and hinder regulatory compliance in healthcare e-commerce.

Existing System — Limitations

→ Basic E-Commerce Only

Traditional online pharmacies focus on browsing and ordering, with limited prescription integration.

→ Manual Prescription Submission

Prescription medicines require manual submission and offline verification, slowing the process significantly.

→ Disconnected Records

Inventory, customer data, prescriptions, and orders are often managed in separate, unintegrated systems.

→ No Role-Based Access

Limited differentiation between customer, pharmacist, and administrator workflows reduces security and accountability.



Proposed System

A secure, integrated pharmacy e-commerce platform with end-to-end prescription validation.



Online Medicine Ordering

Browse, search, and order medicines with a seamless cart and checkout experience.



Prescription Upload

Prescription-only medicines require digital upload before purchase is permitted.



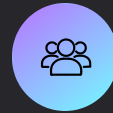
Pharmacist Validation

Pharmacists review and approve prescriptions, allowing orders to proceed only upon verification.



Centralised MySQL

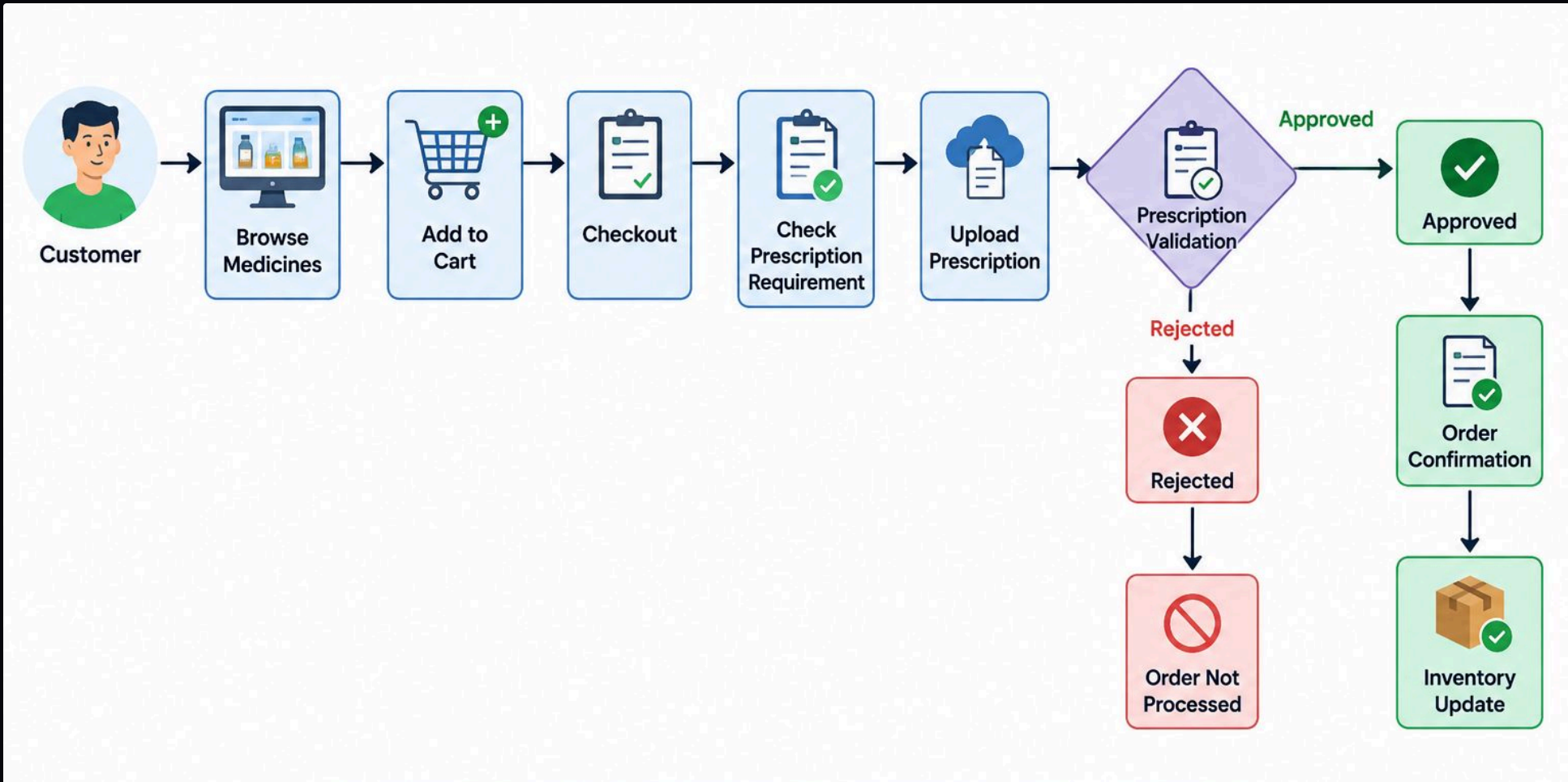
Single database managing customers, medicines, prescriptions, orders, payments, and inventory.



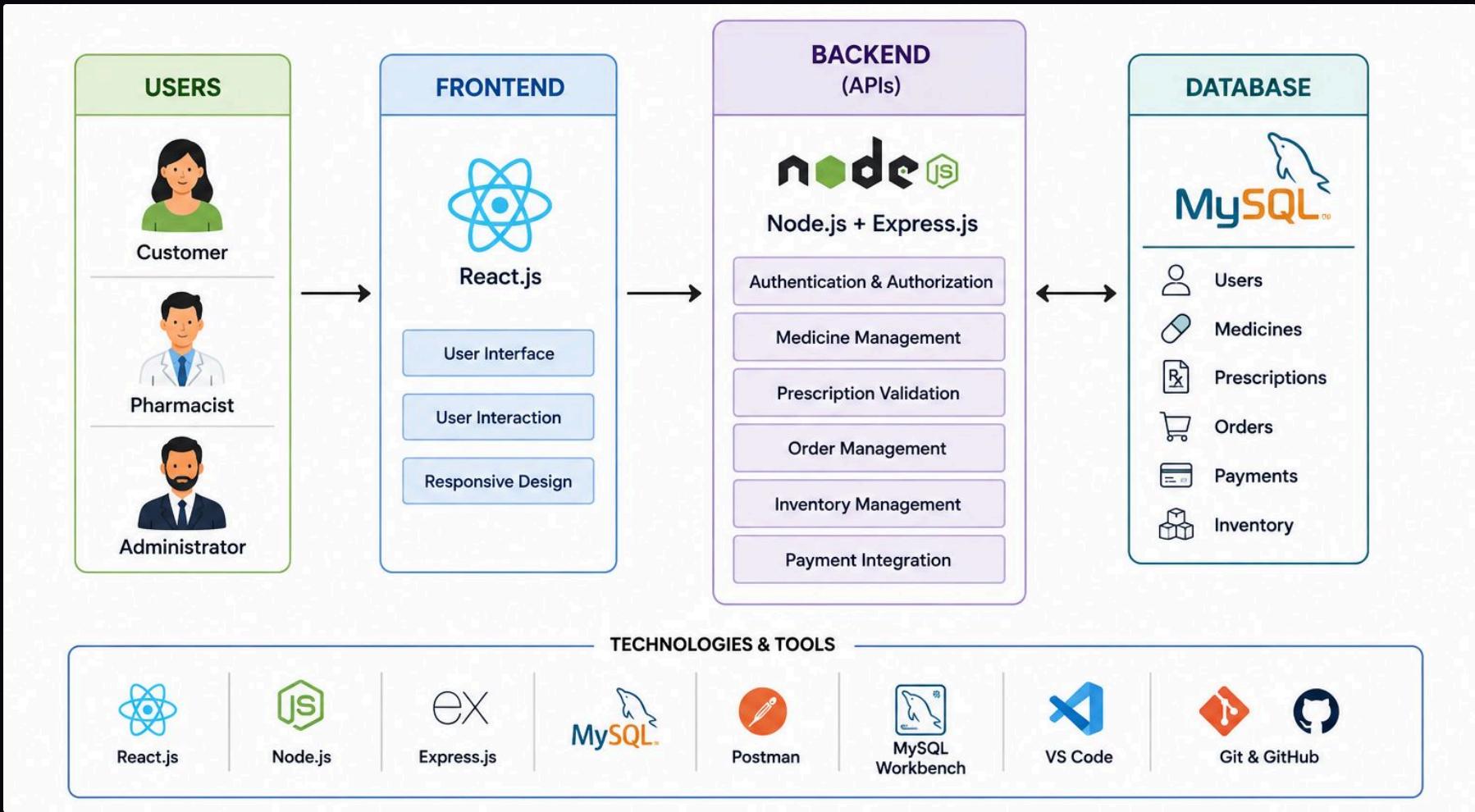
Role-Based Access

Separate portals for Customers, Pharmacists, and Administrators with secure authentication.

System Workflow



System Architecture



Tools & Technologies



Frontend

React.js, HTML, CSS, JavaScript — responsive and dynamic user interface.



Database

MySQL for structured data storage; MySQL Workbench for schema design.



Backend

Node.js with Express.js — RESTful API layer handling business logic.



Development Tools

Visual Studio Code for development; Git & GitHub for version control.



Thank You

We appreciate your time and attention. We welcome your questions and feedback.

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25CS1302E · BATCH 11 · SECTION 2